

Full length article

Withania somnifera dietary supplementation improves lipid profile, intestinal histomorphology in healthy Nile tilapia (*Oreochromis niloticus*), and modulates cytokines response to *Streptococcus* infection

Eman Zahran^{a,*}, Mahmoud G. El Sebaei^{b,f}, Walaa Awadin^c, Samia Elbahnaswy^a, Engy Risha^d, Youssef Elseady^e

^a Department of Internal Medicine, Infectious and Fish Diseases, Faculty of Veterinary Medicine, Mansoura University, Mansoura, 35516, Egypt

^b Department of Biomedical Sciences, College of Veterinary Medicine, King Faisal University, Al-Ahsa, 31982, Saudi Arabia

^c Department of Pathology, Faculty of Veterinary Medicine, Mansoura University, Mansoura, 35516, Egypt

^d Clinical Pathology Department, Faculty of Veterinary Medicine, Mansoura University, Mansoura, 35516, Egypt

^e Department of Physiology, Faculty of Veterinary Medicine, Mansoura University, Mansoura, 35516, Egypt

^f Department of Biochemistry and Chemistry of Nutrition, Faculty of Veterinary Medicine, Mansoura University, Mansoura, 35516, Egypt

ARTICLE INFO

Keywords:

Fish
Herbal plant
Lipid metabolism
Gut morphology
Gene expression
Bacterial challenge

ABSTRACT

Despite *Withania somnifera* (WS), stimulating effects have been investigated on many animal species, its role on lipid profile and intestinal histomorphology in healthy animals, and its modulating role on pro-inflammatory cytokines following infection in fish are yet scarce. In this context, lipid profile, liver, and intestinal histomorphology were measured in Nile tilapia fed with a basal diet or diets containing 2.5 and 5% of supplementary WS for 60 days. Besides, cytokines response was measured at 1, 3, 7, and 14 days following *Streptococcus iniae* (*S. iniae*) infection after the feeding trial. All lipid profile parameters were nominally lowered, excluding **high-density lipoprotein** (HDL) that exhibited a significant increase in WS 5% group compared to other groups. Improved gut health integrity was observed, especially in WS 5% group in terms of increased goblet cell numbers, villous height, the width of lamina propria in all parts of the intestine, and a decrease in the diameter of the intestinal lumen of the distal intestine only. A significant down-regulation in the mRNA transcript level of cytokine genes (interleukin 1 β /IL-1 β , tumor necrosis factor α /TNF α , and interleukin 6/IL-6) was demonstrated in the kidney and spleen of WS-supplemented groups following *S. iniae* infection compared with the control infected (positive control/PC) group. Our findings give new insights for the potential roles of WS dietary inclusion not only on lipid profile and intestinal health integrity improvement in healthy fish under normal rearing but also as a prophylactic against the infection. Thus, WS can be incorporated as a promising nutraceutical in aquaculture.

1. Introduction

Nile tilapia, *Oreochromis niloticus* is one of the most paramount species within tilapias for the sake of its impressive global production performance that is attributed to its rapid growth, adequate survival in different grow-out strategies, and diseases tolerance [1,2]. However, high stocking density in the intensive grow-out system accompanied by poor water quality is considered major stress to fish, which results in increased susceptibility to bacterial pathogens with a severe decrease in the economic gains [3]. In particular, *S. iniae* is one of the most serious bacterial pathogens in finfish aquaculture, causing 30–75% mortality in Nile tilapia [4].

The adoption of various antibiotics to control bacterial diseases in aquaculture gives rise to the emergence of drug-resistant bacteria, immunosuppression, and bioaccumulation of hazardous residues raising a public health concern [5]. In this context, alternative and preventive herbal-based therapeutics are becoming increasingly vital to halt the progression of aquatic diseases. In tilapia aquaculture, many natural medicinal herbal extracts are used in fish as feed supplements to enhance feed utilization, growth performance, antioxidation, and immune response activation [2,6–8].

Withania somnifera (WS) is a herbal plant that belongs to the family Solanaceae, and most pharmacological activities of WS have been attributed to their roots [9]. The main bioactive constituents of

* Corresponding author.

E-mail address: emanzahran@mans.edu.eg (E. Zahran).

<https://doi.org/10.1016/j.fsi.2020.07.056>

Received 29 May 2020; Received in revised form 20 July 2020; Accepted 24 July 2020

Available online 30 July 2020

1050-4648/© 2020 Elsevier Ltd. All rights reserved.

Table 1

Composition and proximate analysis of basal and supplemented diets (% dry matter).

Ingredients	Control	WS 2.5%	WS 5%
Fish meal	25	25	25
Yellow corn	17.65	15.15	12.65
Soya bean meal	20.50	20.50	20.50
Wheat bran	34.0	34.0	34.0
Corn gluten meal	1.0	1.0	1.0
Vegetable oil	0.85	0.85	0.85
Vit and Min mixture	0.50	0.50	0.50
Salt	0.5	0.5	0.5
WS	–	2.5	5
Proximate analysis (% dry matter basis)			
Crude Protein	32.12	31.91	31.70
Crude lipid	6.59	6.50	6.40
Crude fiber	4.65	4.71	4.76
Ash	7.21	7.15	7.43
Ca	0.89	0.87	0.81
*DE (Digestable Energy) Kcal/kg	3067	3012	2977
*Energy/protein ratio	95.48	94.39	93.911

^aTrace minerals & vitamins premixes were prepared to cover the levels of the micro minerals & vitamins for tilapia fish as recommended by NRC [87]. Vitamins premix (IU or mg/kg diet); vit. A 5000, Vit.D3 1000, vit. E 20, vit. k3 2, vit. B1 2, vit. B2 5, vit. B6 1.5, vit. B12 0.02, Pantothenic acid 10, Folic acid 1, Biotin 0.15, Niacin 30. Mineral mixture (mg/kg diet); Fe 40, Mn 80, Cu 4, Zn 50, I 0.5, Co 0.2 & Se 0.2. WS = *Withania Somnifera*. *Calculated values.

Ashwagandha extracts are alkaloids (isopellertierine and anferine), steroidal lactones (withanolides), which withanone, withaferin A, withanolide A, and withanolide D comprise the major fractions, and saponins containing an additional acyl 76 groups (sitoindoside VII and VIII) [10,11]. Therefore, WS is used extensively as a medicinal herbal-drug [12–15], and particularly, the roots are believed to be most potent for therapeutic purposes [16]. Besides, antioxidant properties of WS in stress-exposed animals, concerning its hypolipidemic effect, have been reported in many previous studies [17–22], and also its anti-stress effects in *Oreochromis mossambicus* [23].

The liver and intestines are the most important organs involved in the digestion and absorption process of nutrients from feed. Therefore, their histological analysis, combined with other health parameters, is considered as biomarkers for evaluating the dietary incorporation of natural herbal plants, including WS, on fish health [24].

Cytokines- and chemokines-related genes have been used to monitor the immediate and early stages of fish immune response during infection [25]. Recently, many studies evaluated the anti-inflammatory effects of WS concerning its immunomodulatory role to be developed as a therapeutic for inflammatory diseases [26–29]. Withaferin A and withanolide D have been well studied as a natural agent for suppressing the inflammatory responses via inhibiting IkappaB phosphorylation and the nuclear factor kappa-light-chain-enhancer of activated B cells (NFκB) signaling cascade [30–32], with subsequent inhibition of many pro-inflammatory mediators such as TNF-α, IL-1β, and proteases [29, 33].

As pointed earlier, several research works have been conducted to evaluate WS effects in many animal species exposed to stressors. However, studies concerned the role of dietary WS on the non-stressed animal species with respect mainly to its hypolipidemic effect and its influence on the liver and intestinal morphology; besides, its immunomodulatory role on the expression of the pro-inflammatory cytokines in fish are yet very few. Therefore, the current study was conducted to evaluate the impact of WS root dietary incorporation on lipid profile, liver, and intestines histological indices, besides its immunomodulatory role of pro-inflammatory cytokines to protect against *S. iniae* induced infection in Nile tilapia.

2. Materials and methods

Two successive experiments were carried out in this study. The first experiment investigated the effects of WS supplementation at a level of 2.5 and 5% on lipid profile, intestinal, and liver histopathology of Nile tilapia. The second experiment was conducted to explore the immunomodulatory effects of WS on pro-inflammatory cytokines response following the *S. iniae* infection.

2.1. Fish and herbal diets preparation

Nile tilapia, *Oreochromis niloticus* (average weight 45 g), were obtained from a private fish farm and transferred alive to fish wet lab. The fish were firstly acclimated in an aerated freshwater tank at 24 °C and fed ad lib with a commercial diet (Uccma feed, Egypt) containing 35% crude protein, two times a day for two weeks under (12/12 h light/dark photoperiod) before their use in the study. **Evaluation of overall fish health was carried out during the acclimation period, and only healthy fish, in terms of general appearance and activity level, were used.** WS-supplemented diets were prepared, according to Zahran et al. [22]. The experimental diets were prepared with the locally available ingredients containing 2.5%, and 5% of WS root powder (Table 1). The dried pellets were stored in an air-sealed container and stored in a cool, dry place for further use.

2.2. Challenge experiment

Preparation of *S. iniae* was performed as described in our earlier work [34]. The bacterial concentration was measured by plating the serially diluted bacterial suspension on BHI agar. The estimated concentration was 9.4×10^8 CFU/mL.

2.3. Experimental design

2.3.1. Experiment I

After the acclimation period, fish were divided into three groups (0%, 2.5%, and 5%) in triplicate. Fifteen fish were distributed in each of nine aquarium tanks (80 × 40 × 30 cm) (45 fish/group, $N = 135$). All aquaria were supplied by electric aerators to maintain the oxygen level and underwater internal power filter. Fish were fed at the rate of 3% of body weight twice a day with the respective diets until the end of the experiment, and 50% of water was replaced biweekly over the trial. Water quality parameters (temperature 25–27 °C, dissolved oxygen 6 mg/L, and pH 7.5–8, $\text{NH}_3/\text{NH}_4^+$, and nitrite 0.25 mg/L) were measured via water quality test kits (Aquarium Pharmaceuticals, Inc.). Wastes were siphoned off daily to avoid a higher level of ammonia.

2.3.1.1. Sample collection. After 60 days of feeding, six fish were collected from each group. Fish were euthanized and sampled one at a time using buffered MS-222 (Tricaine methanesulfonate, Finquel®, Argent). Blood samples were then collected via venipuncture using a 3-mL syringe and a 24 GA needle; into a plain centrifuge tube, then centrifuged at $1700 \times g$ for 10 min at 4 °C. The serum was separated carefully and stored at –80 °C until use. The liver and intestine were excised immediately after euthanasia and fixed in 10% buffered formalin for morphometric and histopathological examination. All experimental protocols were approved by the Ethical Research Committee of Mansoura University (Code number R/29).

2.3.1.2. Lipid profile analysis. Serum triglycerides (TG), total cholesterol, low-density lipoprotein (LDL), and HDL were measured spectrophotometrically (BM Co. Germany, 5010) using Spinreact (Spain) commercial kits according to the manufacturer's instructions.

2.3.1.3. Morphometric and histopathological examination. Tissue

Table 2

Primers used for real-time PCR analysis of gene expression.

Gene	Forward primer (5'-3')	Reverse primer (5'-3')	Accession number
IL-1 β	CTGTGTGACAACTCTCAGAAAGATCTTG	TGTCATCATGGTATTGCTCCAGAA	Zahran et al. [88]
TNF- α	GAATACAAGGCCAGAAAGGATGACAC	GACCTTTTGAAGTCGCTGCCTTC	
IL-6	ACAGAGGAGGCGGAGATG	GCAGTGCTTCGGGATAGAG	Elbahnasw and Elshopakey [89]
B-Actin	CAGCAAGCAGGAGTACGATGAG	TGTGTGGTGTGTGGTTGTTT	

specimens were collected from 6 fish per group, including liver and the three parts of the intestine, the proximal part (after the pyloric part of the stomach to before the spiral part of the intestines), the middle part (the spiral part of the intestines) and distal part (after spiral part of the intestines to 2 cm before anal opening), and fixed in 10% buffered formalin. Subsequently, the specimens were dehydrated in a graded ethanol series and then embedded in paraffin wax. All sections were cut at four μ m and stained with hematoxylin and eosin (H&E). Images of the liver and intestine were captured with an Olympus BX41 microscope supplied with a DVC 1300C color digital camera.

Histometric measurements were applied to tissue sections from the liver and intestine using image analysis (image J = <http://imagej.en.softonic.com>). Measurements in the intestine were as follows: a) diameter of the intestinal lumen, b) the number of the mucosal folds per section, c) mucosal folds height (from top to the base of the mucosal fold), d) width (the maximum width of the fold along a line perpendicular to the fold height), e) perimeter, f) area (area within the perimeter for each fold), g) thickness of the muscular layer and h) goblet cells number [24]. Values of (c-h) used for analysis were the means of the highest five-folds per section. Partly or wholly fused, two folds were treated as one. Liver measurements included a cytoplasmic/nuclear ratio (estimated in 50 hepatocytes per section) [35].

2.3.2. Experiment II

Fish (39 fish/group, $N = 117$) remaining, from the experiment I, were fed on the basal diet for one week. Then all groups were injected I/P with 0.1 mL of PBS containing *S. iniae* at 9.4×10^8 CFU/mL. At days 1, 3, 7, and 14 post-infection, six fish ($n = 6$ /group) were sampled from the control infected and WS-infected groups and euthanized with a buffered MS-222 one at a time in a separate container to collect spleen and head kidney tissue samples for gene expression analysis.

2.3.2.1. Cytokine gene expression analysis. Total RNA was isolated from the head kidney and spleen of the Nile tilapia at 1, 3, 7, and 14 days post-infection using Qiagen RNeasy® kit according to the manufacturer's instructions. cDNA was synthesized using RevertAid™ Reverse Transcriptase (Fermentas), according to the manufacturer's instructions. Real-time PCR was carried out using SYBR Green PCR Master Mix (Fermentas, USA). Each reaction was performed in a volume of 25 μ L. The amplification program set at 95 °C for 10 min, followed by 40 cycles of denaturation at 95 °C for 15 s, annealing at 60 °C for 30 s and extension at 72 °C for 30 s, using a LightCycler® 480 Real-Time PCR System (Roche). The expression of pro-inflammatory cytokines IL-1 β , TNF- α , and IL-6 was examined. The primers for qPCR are detailed in Table 2. A fold change was calculated by comparison of the gene expression level (normalized to the reference gene β -actin) between the

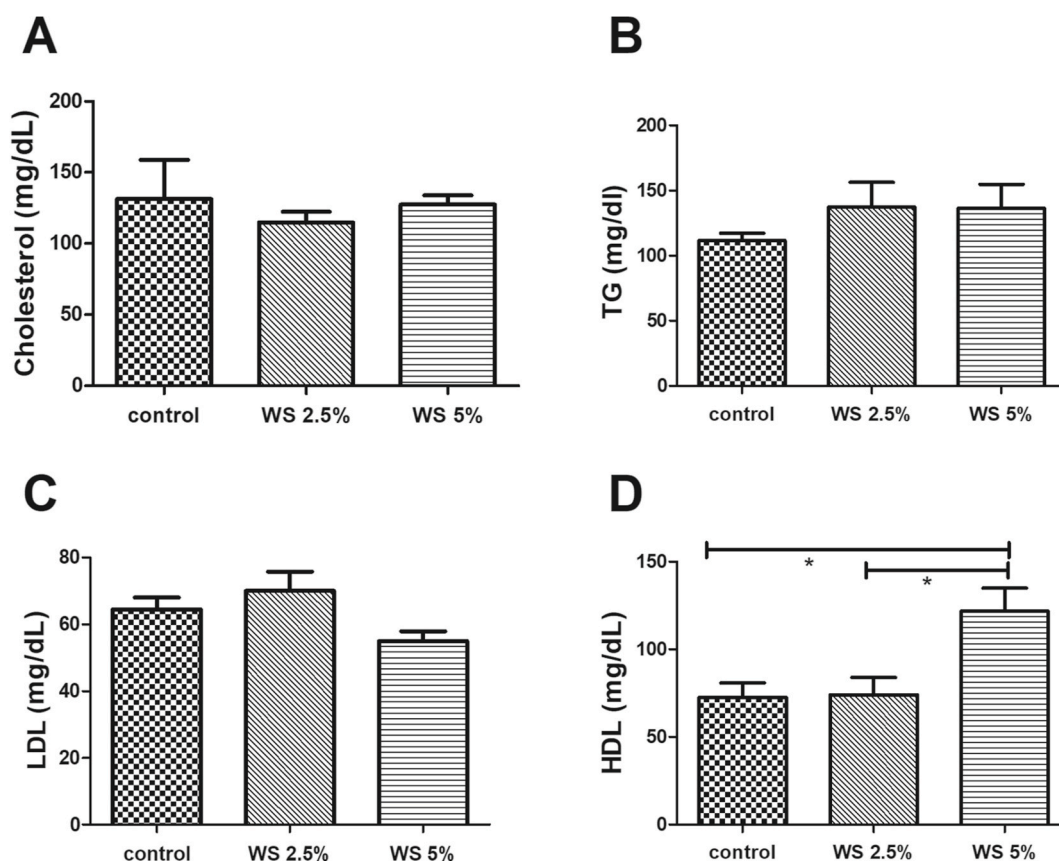


Fig. 1. Lipid profile (A) cholesterol, (B) TG, (C) LDL, and (D) HDL in Nile tilapia fed 0%, 2.5%, 5% of WS supplemented diets. Data were expressed as the mean of six fish $\pm$ SEM. Values with a different letter superscript are significantly different between groups (ANOVA, $*P < 0.05$).

Table 3

Histomorphometric analysis of the proximal, middle and distal intestine of Nile tilapia fed 0%, 2.5%, 5% of WS supplemented diets.

Intestinal parts	Parameters	Control	2.5%	5%
Proximal	Diameter of lumen	1915 ± 110.5	2082 ± 152.4	2081 ± 177.4
	Number of mucosal folds	25.67 ± 1.85	25.90 ± 3.59	39.67 ± 6.66
	Number of goblet cells	15.36 ± 1.14 ^b	20.80 ± 2.02 ^a	22.20 ± 1.24 ^a
	Height	680.8 ± 26.92	674.4 ± 12.51	679.3 ± 14.01
	Perimeter	1576 ± 81.04	1710 ± 48.43	1841 ± 192.8
	Area	3427 ± 167.3	3678 ± 106.2	3294 ± 97.55
	Width of lamina propria	72.8 ± 5.22 ^b	90.77 ± 3.9 ^a	95.82 ± 3.42 ^a
Middle	Thickness of the muscle layers	125.6 ± 5.54 ^a	105.3 ± 5.62 ^b	90.61 ± 2.57 ^c
	Diameter of lumen	1289 ± 83.21	2443 ± 33.8	1323 ± 66.36
	Number of mucosal folds	31 ± 1	18.67 ± 4.08	21.60 ± 2.92
	Number of goblet cells	13.75 ± 1.18 ^b	12.81 ± 1.09 ^b	18.56 ± 1.56 ^a
	Height	423.6 ± 13.24 ^b	424.4 ± 13.32 ^b	646.8 ± 70.90 ^a
	Perimeter	937.6 ± 35.50	1051 ± 35.70	1046 ± 40.53
	Area	2003 ± 85.42	2311 ± 72.18	2253 ± 88.77
Distal	Width of lamina propria	54.60 ± 1.94 ^b	90.83 ± 5.31 ^a	113.3 ± 7.20 ^a
	Thickness of muscle	112 ± 7.45	115.4 ± 9.18	99.15 ± 4.64
	Diameter of lumen	2312 ± 199.8 ^a	1974 ± 174.2 ^{ac}	1395 ± 77.8 ^{bc}
	Number of mucosal folds	24.33 ± 2.6	17.75 ± 6.3	10 ± 3
	Number of goblet cells	9.26 ± 1.56	10.43 ± 1.15	10.64 ± 1.05
	Height	227.1 ± 14.95 ^a	252 ± 12.87 ^{ac}	300 ± 28.63 ^{bc}
	Perimeter	535.3 ± 27.58 ^b	636.4 ± 29.91 ^{ab}	761.3 ± 69.56 ^a
	Area	1142 ± 63.85 ^b	1406 ± 70.93 ^b	1552 ± 143.5 ^a
	Width of lamina propria	70.22 ± 8.38 ^{bc}	105.9 ± 16.99 ^c	120.4 ± 8.99 ^{ac}
	Thickness of muscle	74.26 ± 3.93	75.38 ± 6.56	101.8 ± 14

control infected group (PC) and the WS-infected groups at each time point [36].

2.4. Statistical analysis

Data were subjected to normality and homoscedasticity checks by the Kolmogorov-Smirnov and Levene tests, respectively. Data were then analyzed using the SPSS software statistical package (SPSS version 20.0 for Window) and are expressed as mean ± standard error (SE). Statistical significance was determined using one-way ANOVA followed by Duncan's multiple range test, and values with $P < 0.05$ were considered statistically significant.

3. Results

3.1. Lipid profile analysis

WS supplementation at 5% resulted in a significant increase in HDL level compared to other groups. Meanwhile, no statistical changes were noticed on other lipid profile parameters in WS-supplemented groups compared to the control one (Fig. 1).

3.2. Histomorphometric analysis

Microscopical examination of intestine and liver from WS-supplemented diets did not reveal any histopathological alterations and appeared similar to that of the control group Fig. 3. Histomorphometric measurements were presented in Table 3. Nile tilapia fed 2.5 and 5% WS showed a significant increase in the number of goblet cells and width of lamina propria, accompanied by a significant decrease in thickness of the muscle layers in the proximal intestine compared with the control group. In the middle intestine, both groups showed a significant increase in the width of lamina propria compared with the control group, while only Nile tilapia fed 5% WS demonstrated a significant increase in the goblet cell numbers and the height of intestinal villi compared with 2.5% WS and control groups. In the distal intestine, the diameter of the lumen, villous height, and perimeter were significantly increased in Nile tilapia fed 5% WS compared to the control group with no statistical changes between both WS-supplemented groups. Meanwhile, the villous

area was significantly different in Nile tilapia fed 5% WS compared to other groups. Hepatocyte cytoplasmic/nuclear ratio was nominally decreased in Nile tilapia fed 2.5 and 5% WS compared to the control group (Fig. 2).

3.3. Cytokine gene expression analysis

The prophylactic effect of WS on the expression of the cytokine genes following *S. iniae* infection at days 1, 3, 7, and 14 are presented in Fig. 4. WS-supplementation elicited different modulation of the cytokines' response, where the mRNA expression levels of TNF- α and IL-1 β showed a significant down-regulation in both tissues at both WS supplemented levels (2.5 and 5%) along with significant variations in their expression levels overtime points. IL-6 was significantly downregulated in WS 2.5% group in the spleen from day 3, but an earlier response in the kidney was noticed from day 1, up to day 14 post-infection compared to the PC group. In WS 5% group, spleen IL-6 showed a significant down-regulation overtime points compared to the PC group, while the mRNA level of IL-6 in head kidney downregulated later at 3–14 days post-infection compared to the PC group.

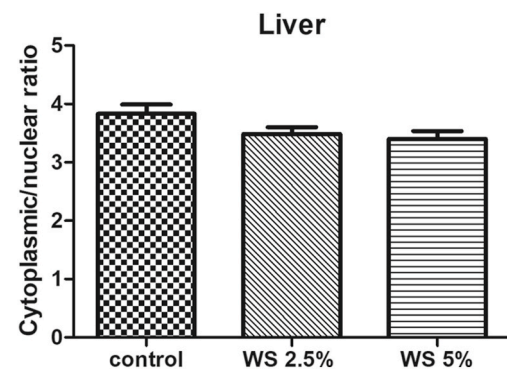


Fig. 2. Hepatocytes cytoplasmic/nuclear ratio in Nile tilapia fed 0%, 2.5%, 5% of WS supplemented diets. Data were expressed as the mean of six fish ± SEM. Values with a different letter superscript are significantly different between groups (ANOVA, $P < 0.05$).

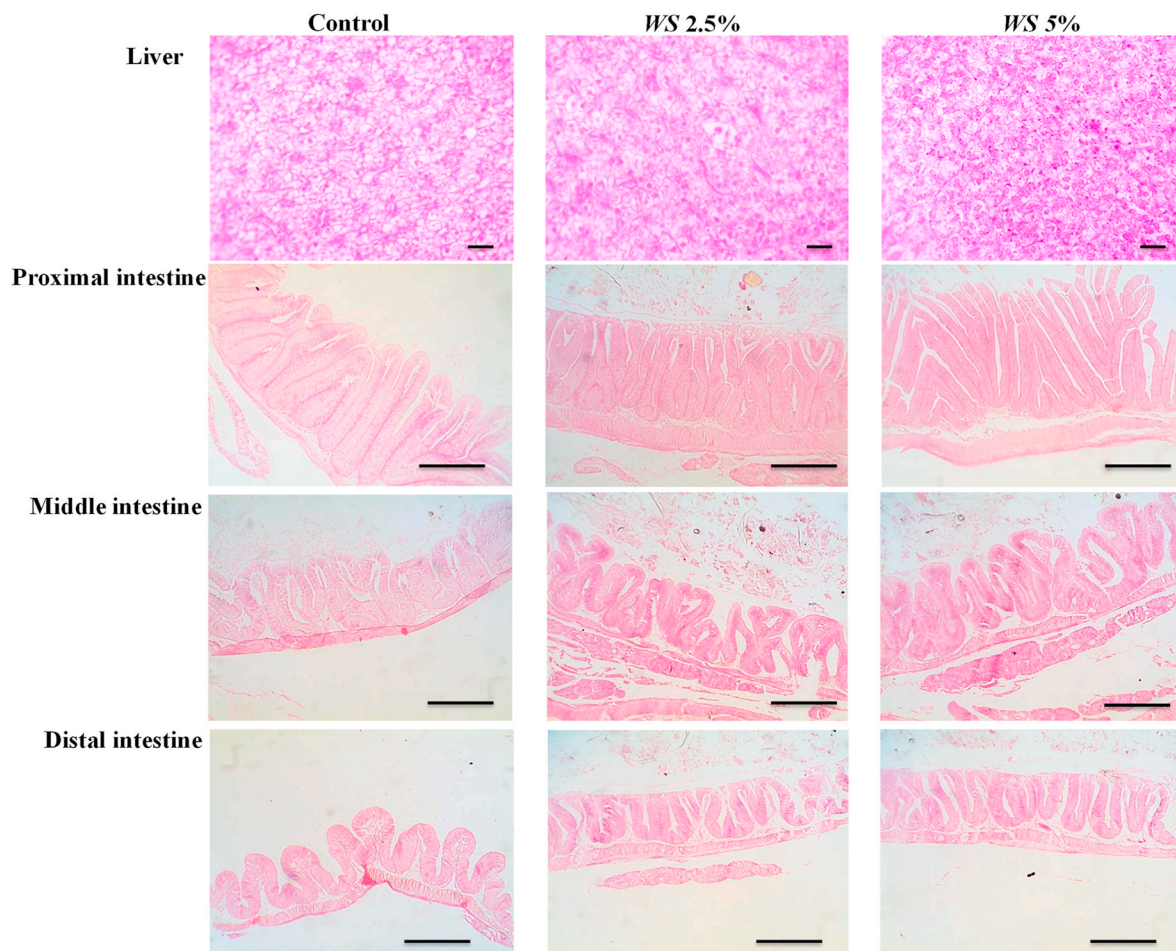


Fig. 3. Liver and intestine light microscopy of Nile tilapia fed 0%, 2.5%, 5% of WS supplemented diets. Liver (X: 400 bar 50), proximal, middle, and distal intestine (X: 100 bar 100).

4. Discussion

Lipid metabolism regulation by WS root extract, including hypolipidemic effects over blood biochemistry, was documented in rats and broiler chicks [18,37–39]. As observed, WS had no side effects on the lipid profile parameters. However, it showed a favorable effect, particularly at the level of 5%, where a significant increase in HDL was noticed compared to other groups. These observations might be justified in the view of the hypolipidemic activity of WS owing to its high flavonoid content with other compounds like withanolides, withaferin-A, and alkaloids in root powder [10,21,40,41]. A similar trend was found in Mozambique tilapia fed WS (1%) for 45 days, where levels of triglycerides and total cholesterol decreased than the control fish [23]. Similar effects were observed in black rockfish, where tea extract dietary supplementation (5%) led to a decline in plasma total cholesterol, with no significance on plasma HDL cholesterol [42].

Total cholesterol, triacylglycerol, and LDL were significantly reduced in Nile tilapia fed 0.5, 2, and 4% of *Aloe vera* for eight weeks compared to the control group [43]. Besides, a notable decrease in total lipids, triglycerides and cholesterol levels in Asian sea bass fed garlic diets at 5, 10, 15, and 20 g/kg feed for two weeks compared with the control [44]. Other similar studies were documented in olive flounder, and gilthead sea bream fed green and white tea supplemented diets at 5 and 2.9% for 7 and 30 days, respectively [45,46]. Lipid peroxidation and oxidative stress caused by excessive reactive oxygen species (ROS) can be voided by increased HDL activity, which in turn inhibit LDL oxidation and its subsequent toxic effects after oxidation, and eventually lowering the plasma lipids [43,47,48]. Thus, however, no statistical changes were

noted excluding HDL, our findings can prove the safety of incorporation WS as a functional feed in Nile tilapia, which can protect in cases of lipid peroxidation in terms of HDL finding.

Diet composition can influence the intestinal structure and, consequently, its absorptive capacity and animal performance [49]. As observed, Nile tilapia fed 5% WS had significantly increased goblet cell numbers in the proximal and middle part, increased villous height in middle and distal parts, and increased width of lamina propria in all parts of the intestine. However, a decrease in the diameter of the intestinal lumen was noted in the distal intestine only. These findings suggested that the WS supplementation was able to increase the absorptive surface of the intestine and thus improve the nutrient digestibility. The increased number of mucosal folds and villous height contribute to efficient absorptive capacity as well [50–52]. Increased goblet cell numbers of the intestine play an important role in safeguarding the intestinal lining [53,54], protecting the mucosal surface and trapping pathogenic bacteria and parasites owing to the increased mucus production [55].

Nile tilapia fed 5% WS demonstrated a significant decrease in the thickness of the muscle layers of proximal intestine only, indicating increased peristalsis. Herbivorous fish need less rectal muscle contraction than the carnivorous ones since they consumed high fibrous contents which encourage the intestinal peristalsis [56]. This layer promotes intestinal motility, which accompanied by enhanced efficiency of mucosal folds [57]. As observed, a nominal decrease in hepatocyte cytoplasmic/nuclear ratio was noticed in Nile tilapia fed WS-supplemented diets compared to the control, suggesting that WS up to 5% did not lead to glycogen deposition in hepatocytes. Our findings

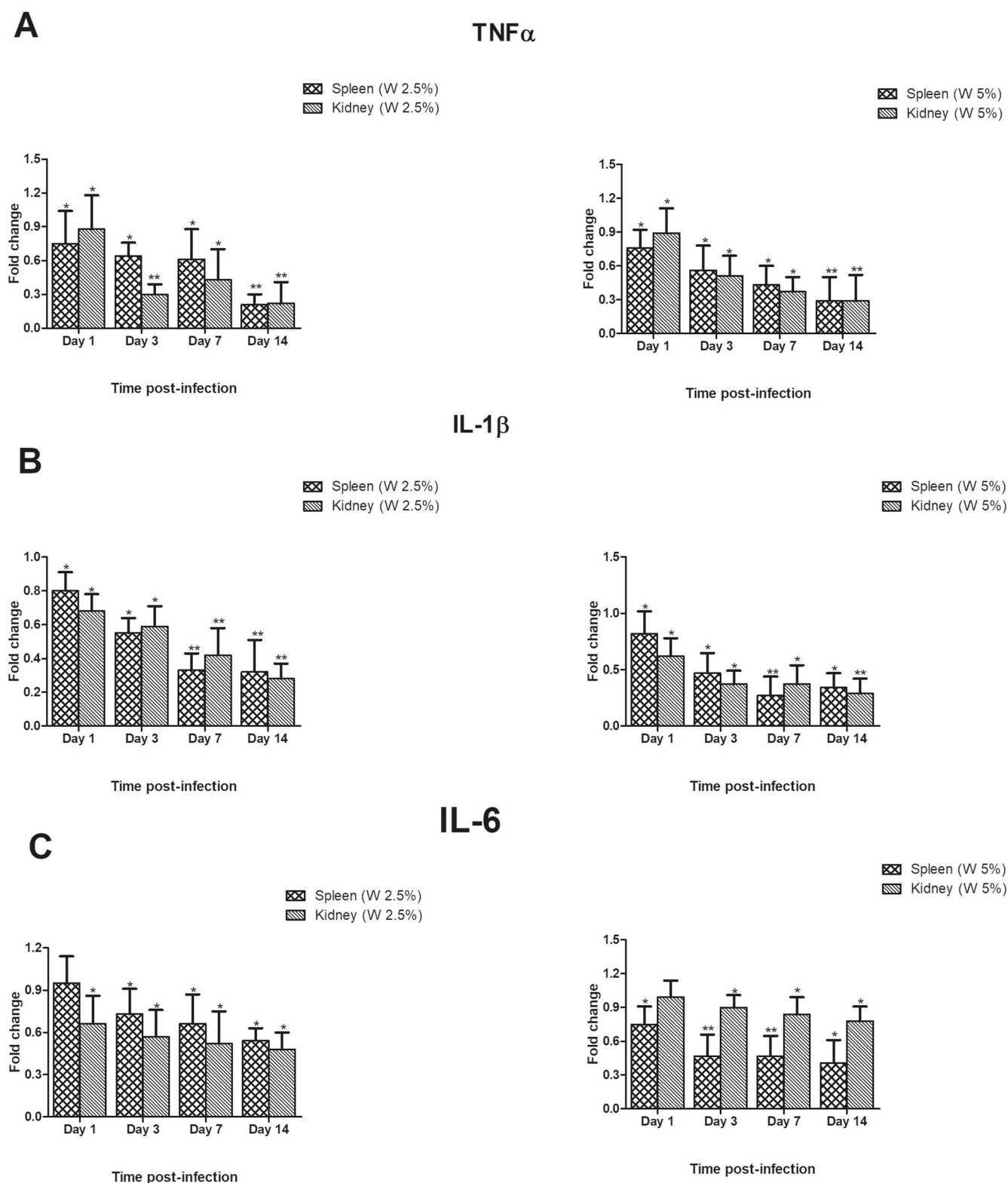


Fig. 4. The mRNA transcriptional levels of (A) TNF- α , (B) IL-1 β and (C) IL-6 (as measured by qPCR) in the head-kidney and spleen of Nile tilapia fed 0%, 2.5%, 5% of WS supplemented diets. Data were expressed as the mean of six fish $\pm$ SEM. Values with asterisk (*) superscript are significantly different from control (ANOVA, * P < 0.05, ** P < 0.01).

were in line with previous studies considering the hepatocyte cytoplasmic/nuclear ratio as a useful indicator in nutritional studies [58–60].

These findings are consistent with previous studies concerned herbal dietary supplementation such as Nile tilapia fed can *Astragalus*

polysaccharides at levels of 1500 mg/kg [8]; the gibel carp and Caspian kutum fingerlings fed dietary chitosan [61,62]; European sea bass and rainbow trout fed mannan oligosaccharide for eight weeks and 60 days, respectively, with a positive effect on the gut epithelial integrity [63, 64]; and in rainbow trout fed diets supplemented with *Aloe vera* [65].

These studies support our findings and prove that dietary supplementation of WS up to 5% diet inclusion positively influenced the gut integrity and functionality.

Research studies on WS concerned the expression of the immune-related genes, specifically, pro-inflammatory cytokines, including IL-1 β , TNF- α , and IL-6 post-infection, are yet few. Therefore, in this study, WS prophylactic efficacy against inflammatory response caused by *S. iniae* challenge was investigated. It has been contended that the non-specific fish immunity lacks memory and therefore, the extent of immunomodulator induced responses will be less than the specific one [66,67]. However, WS root powder proved its efficacy as a prophylactic pretreatment of animals preceding the induction of disease [29,68] and also in various disease animals models [32,69].

TNF- α is a potent pro-inflammatory cytokine associated with antibacterial, anti-parasitic, and antiviral immunity in fish [70]. As well, it is considered a synchronized pro-inflammatory sign for the expression of chemokines and adhesion molecules that have either diverse critical aspects on an innate immunity or severe damage to the target organs [71]. It is activated and regulated during infections and inflammatory disorders via ROS, which in turn stimulates the intracellular signaling pathway NF- κ B [72].

In the current study, the mRNA expression levels of TNF- α and IL-1 β were in a similar pattern, where a significant down-regulation was noticed in both tissues at both WS supplemented levels. These results might be attributed to withaferin A, an active component in the root extract of WS, which is a potent inhibitor of the pro-inflammatory transcription factors NF- κ B signaling pathway and so inhibit the TNF- α activation as previously reported [30,39]. WS significantly inhibited the expression of TNF- α in the stainless steel implant triggered inflammation in zebrafish [27]. In the same line, formalin-killed atypical *Aeromonas salmonicida* induced a significant decrease in TNF- α 2 mRNA expression in *Oncorhynchus mykiss* fed the *Aloe vera* supplemented diet for six weeks [73]. Similar to our findings, lower serum levels of TNF- α were also observed in mice fed with WS extract for 5 and 10 days [74].

Interleukin-1 β is another pro-inflammatory cytokine induces a cascade of early responses leading to inflammation. Also, it is responsible for phagocytosis stimulation, lymphocyte proliferation, superoxide production, and enhances the release of other cytokines [75,76]. Our finding is consistent with Zanuzzo et al. [73], where IL-1 β down-regulated in steelhead trout (*Oncorhynchus mykiss*) fed a diet containing 0.5% of *Aloe vera* for six weeks after 3h of injection with *Aeromonas salmonicida*; and in the head kidney of rainbow trout fed dietary *Lactobacillus plantarum* CLFP 3 after *Lactococcus garvieae* infection [46].

IL-6 cytokine is also known to play important roles in the inflammatory response [77]. It rapidly induced upon bacterial invasion [78,79]. Besides, IL-6 plays an important role in T cells and B cell functions [80]. Our findings demonstrate that both 2.5% and 5% of WS dietary supplementation had a significant effect on downregulating IL-6 mRNA expression level following the *S. iniae* challenge. Interestingly, WS supplementation significantly suppressed the serum levels of TNF- α , IL-1 β , IL-6 cytokines, and protein levels; besides, inhibited tissue expression mRNA level of TNF- α and IL-1 β in the pyriform cortex, hippocampus, and hypothalamus regions of albino rat brain after treatment with a systemic lipopolysaccharide (LPS)-induced neuroinflammation [29].

As observed, WS immunomodulatory role through the down-regulation of the expression levels of IL-1 β , TNF- α , and IL-6 cytokines during the bacterial infection might be attributed to its major anti-inflammatory bioactive constituent, including sitoindosides VII-X and withaferin A (glycowithanolides) [39,81–83]. Besides, its content of phenolic acids and flavonoids block the transcriptional signaling factors NF- κ B, activating protein-1 (AP-1), and a nuclear factor-erythroid 2-related factor 2 (Nrf2) [27,84]. Interestingly, the ability of WS to modulate the lipid peroxidation and oxidative damage in term of increased the HDL herein may contribute to regulate critical mediators in inflammatory events, such as the challenge with *S. iniae* and suppress the cytokine expression levels in the head kidney and spleen in Nile

tilapia fed WS supplemented diets [85,86].

5. Conclusion

In summary, the findings documented herein indicated that WS dietary supplementation has significantly improving effects on lipid profile and histomorphometric indices of liver and intestine. Therefore, it can be incorporated as a safe, functional feed in the fish diet. Furthermore, WS exhibited a prophylactic effect by modulating the induced expression of pro-inflammatory cytokines during the infections. Thus, WS can be considered as a promising nutraceutical in aquaculture.

CRedit authorship contribution statement

Eman Zahran: Conceptualization, Methodology, Investigation, Validation, Supervision, Writing - review & editing. **Mahmoud G. El Sebaei:** Methodology, Investigation, Visualization, Writing - original draft. **Walaa Awadin:** Methodology, Investigation, Visualization, Writing - original draft. **Samia Elbahnaswy:** Writing - original draft. **Engy Risha:** Methodology, Investigation, Visualization. **Youssef Elseady:** Methodology, Investigation, Visualization.

Acknowledgments

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

References

- [1] N.F. Soliman, D.M. Yacout, Aquaculture in Egypt: status, constraints and potentials, *Aquacult. Int.* 24 (5) (2016) 1201–1227.
- [2] N.N. Gabriel, Review on the progress in the role of herbal extracts in tilapia culture, *Cogent Food Agric.* 5 (1) (2019) 1619651.
- [3] Y.-r. Wu, Q.-f. Gong, H. Fang, W.-w. Liang, M. Chen, R.-j. He, Effect of *Sophora flavescens* on non-specific immune response of tilapia (GIFT *Oreochromis niloticus*) and disease resistance against *Streptococcus agalactiae*, *Fish Shellfish Immunol.* 34 (1) (2013) 220–227.
- [4] C.A. Shoemaker, B.R. LaFrentz, P.H. Klesius, J.J. Evans, Protection against heterologous *Streptococcus iniae* isolates using a modified bacterin vaccine in Nile tilapia, *Oreochromis niloticus* (L.), *J. Fish. Dis.* 33 (7) (2010) 537–544.
- [5] C. Bulfon, D. Volpatti, M. Galeotti, Current research on the use of plant-derived products in farmed fish, *Aquacult. Res.* 46 (3) (2015) 513–551.
- [6] T. Citarasu, Herbal biomedicines: a new opportunity for aquaculture industry, *Aquacult. Int.* 18 (3) (2010) 403–414.
- [7] M.H. Ahmad, M. Abdel-Tawwab, The use of caraway seed meal as a feed additive in fish diets: growth performance, feed utilization, and whole-body composition of Nile tilapia, *Oreochromis niloticus* (L.) fingerlings, *Aquaculture* 314 (1–4) (2011) 110–114.
- [8] E. Zahran, E. Risha, F. AbdelHamid, H.A. Mahgoub, T. Ibrahim, Effects of dietary *Astragalus polysaccharides* (APS) on growth performance, immunological parameters, digestive enzymes, and intestinal morphology of Nile tilapia (*Oreochromis niloticus*), *J. Fish Shellfish Immunol.* 38 (1) (2014) 149–157.
- [9] F. Malik, J. Singh, A. Khajuria, K.A. Suri, N.K. Satti, S. Singh, M.K. Kaul, A. Kumar, A. Bhatia, G.N. Qazi, A standardized root extract of *Withania somnifera* and its major constituent withanolide-A elicit humoral and cell-mediated immune responses by up regulation of Th1-dominant polarization in BALB/c mice, *Life Sci.* 80 (16) (2007) 1525–1538.
- [10] G.L. Gupta, A. Rana, M.A.G. Phcog, Plant review *Withania somnifera* (Ashwagandha): a review, *Pharmacogn. Rev.* 1 (1) (2007) 129–136.
- [11] M.H. Mirjalili, E. Moyano, M. Bonfill, R.M. Cusido, J. Palazón, Steroidal lactones from *Withania somnifera*, an ancient plant for novel medicine, *Molecules* 14 (7) (2009) 2373–2393.
- [12] V. Kumar, A. Dey, M.B. Hadimani, T. Marcović, M. Emerald, Chemistry and pharmacology of *Withania somnifera*: an update, *TANG* 5 (1) (2015) 1–13.
- [13] K. Girish, K. Machiah, S. Ushanandini, K. Harish Kumar, S. Nagaraju, M. Govindappa, M. Vedavathi, K. Kemparaju, Antimicrobial properties of a non-toxic glycoprotein (WSG) from *Withania somnifera* (Ashwagandha), *J. Basic Microbiol.* 46 (5) (2006) 365–374.
- [14] F. Malik, A. Kumar, S. Bhushan, D.M. Mondhe, H.C. Pal, R. Sharma, A. Khajuria, S. Singh, G. Singh, A.K. Saxena, Immune modulation and apoptosis induction: two sides of antitumoural activity of a standardised herbal formulation of *Withania somnifera*, *Eur. J. Cancer* 45 (8) (2009) 1494–1509.
- [15] M. Ven Murthy, P. K. Ranjekar, C. Ramassamy, M. Deshpande, Scientific basis for the use of Indian ayurvedic medicinal plants in the treatment of neurodegenerative disorders: 1. Ashwagandha, *Central Nervous Sys. Agents Med. Chem. (Form. Curr. Med. Chem. Central Nervous Sys. Agents)* 10 (3) (2010) 238–246.

- [16] U. Minhas, R. Minz, A. Bhatnagar, Prophylactic effect of *Withania somnifera* on inflammation in a non-autoimmune prone murine model of lupus, *J. Drug Discoveries Therap. Clin. Risk Manage.* 5 (4) (2011) 195–201.
- [17] J.N. Dhuley, Effect of *ashwagandha* on lipid peroxidation in stress-induced animals, *J. Ethnopharmacol.* 60 (2) (1998) 173–178.
- [18] N.P. Visavadiya, A. Narasimhacharya, Hypocholesteremic and antioxidant effects of *Withania somnifera* (Dunal) in hypercholesteremic rats, *Phytomedicine* 14 (2–3) (2007) 136–142.
- [19] S. Khan, F. Malik, K.A. Suri, J. Singh, Molecular insight into the immune up-regulatory properties of the leaf extract of *Ashwagandha* and identification of Th1 immunostimulatory chemical entity, *Vaccine* 27 (43) (2009) 6080–6087.
- [20] N. Singh, M. Bhalla, P. de Jager, M. Gilca, An overview on *ashwagandha*: A Rasayana (Rejuvenator) of Ayurveda, *Afr. J. Tradit., Complementary Altern. Med.* 8 (5S) (2011).
- [21] J. Ansari, S.H. Khan, A.U. Haq, T. Ahmad, M.I. Abbass, Effect of supplementation of *Withania somnifera* (Linn.) dunal roots on growth performance, serum biochemistry, blood hematology, and immunity of broiler chicks, *J. Herbs Spices Med. Plants* 19 (2) (2013) 144–158.
- [22] E. Zahran, E.A.A. El-Gawad, E. Risha, Dietary *Withania somnifera* root confers protective and immunotherapeutic effects against *Aeromonas hydrophila* infection in Nile tilapia (*Oreochromis niloticus*), *Fish Shellfish Immunol.* 80 (2018) 641–650.
- [23] G. Immanuel, R. Uma, P. Iyapparaj, T. Citarasu, S. Punitha Peter, M. Michael Babu, A. Palavesam, Dietary medicinal plant extracts improve growth, immune activity and survival of tilapia *Oreochromis mossambicus*, *J. Fish. Biol.* 74 (7) (2009) 1462–1475.
- [24] B.S. Rašković, M.B. Stanković, Z.Z. Marković, V.D. Poleksić, Histological methods in the assessment of different feed effects on liver and intestine of fish, *J. Agric. Sci.* 56 (1) (2011) 87–100.
- [25] H.G. Abo-Al-El, An introduction to selected innate immune-relevant genes in fish, *Appl. Ecol. Environ. Res.* 16 (2) (2018) 955–976.
- [26] M. Rasool, P. Varalakshmi, Immunomodulatory role of *Withania somnifera* root powder on experimental induced inflammation: an in vivo and in vitro study, *Vascul. Pharmacol.* 44 (6) (2006) 406–410.
- [27] S. Sivamani, B. Joseph, B. Kar, Anti-inflammatory activity of *Withania somnifera* leaf extract in stainless steel implant induced inflammation in adult zebrafish, *J. Genetic Eng. Biotechnol.* 12 (1) (2014) 1–6.
- [28] G.Y. Sun, R. Li, J. Cui, M. Hannink, Z. Gu, K.L. Fritsche, D.B. Lubahn, A. Simonyi, *Withania somnifera* and its withanolides attenuate oxidative and inflammatory responses and up-regulate antioxidant responses in BV-2 microglial cells, *NeuroMolecular Med.* 18 (3) (2016) 241–252.
- [29] M. Gupta, G. Kaur, *Withania somnifera* as a potential anxiolytic and anti-inflammatory candidate against systemic lipopolysaccharide-induced Neuroinflammation, *NeuroMolecular Med.* 20 (3) (2018) 343–362.
- [30] M. Kailash, W.V. Berghe, A. Heyerick, J. Horion, J. Piette, C. Libert, D. De Keukeleire, T. Essawi, G. Haegeman, Withaferin A strongly elicits I κ B kinase β hyperphosphorylation concomitant with potent inhibition of its kinase activity, *J. Biol. Chem.* 282 (7) (2007) 4253–4264.
- [31] R. Maitra, M.A. Porter, S. Huang, B.P. Gilmour, Inhibition of NF κ B by the natural product Withaferin A in cellular models of Cystic Fibrosis inflammation, *J. Inflammation* 6 (1) (2009) 15.
- [32] K. Heynink, M. Lahtela-Kakkonen, P. Van der Veken, G. Haegeman, W.V. Berghe, Withaferin A inhibits NF-kappaB activation by targeting cysteine 179 in IKK β , *Biochem. Pharmacol.* 91 (4) (2014) 501–509.
- [33] M. Gupta, G. Kaur, Aqueous extract from the *Withania somnifera* leaves as a potential anti-neuroinflammatory agent: a mechanistic study, *J. Neuroinflammation* 13 (1) (2016) 193.
- [34] E. Zahran, E. Risha, S. Elbahnaswy, H.A. Mahgoub, A.A. El-Moaty, Tilapia piscidin 4 (TP4) enhances immune response, antioxidant activity, intestinal health and protection against *Streptococcus iniae* infection in Nile tilapia, *Aquaculture* 513 (2019) 734451.
- [35] Z. Marković, V. Poleksić, Z. Dulić, M. Stanković, Carp production intensification in traditional semi-intensive culture systems by application of extruded feed and selected fish fry, *J. Aquacult.* (2009) 14–17.
- [36] T. Wang, B. Gorgoglione, T. Maehr, J.W. Holland, J.L.G. Vecino, S. Wadsworth, C. J. Secombes, Fish suppressors of cytokine signaling (SOCS): gene discovery, modulation of expression and function, *J. Signal Transduct.* 2011 (2011).
- [37] Q. Uddin, L. Samiulla, V. Singh, S. Jamil, Phytochemical and pharmacological profile of *Withania somnifera* Dunal: a review, *J. Appl. Pharmaceut. Sci.* 2 (2012) 170–175, 01.
- [38] Z. Rehman, S. Khan, N. Chand, K. Khan, R. Khan, M. Arshad, A. Akhtar, A. Tanveer, Association of water based infusion of *allium sativum* and *withania somnifera* with serum lipid profile of broiler chicks, *Sarhad J. Agric.* 29 (2) (2013) 249–254.
- [39] N. Alam, M. Hossain, M.I. Khalil, M. Moniruzzaman, S.A. Sulaiman, S.H. Gan, Recent advances in elucidating the biological properties of *Withania somnifera* and its potential role in health benefits, *Phytochem. Rev.* 11 (1) (2012) 97–112.
- [40] F. Khursheed, K. Rehman, M. Akhtar, M. Dogar, B. Khalil, Comparative antilipidemic effects of native and gemmo-treated *Withania somnifera* (Asghand) extracts, *J. Appl. Pharmaceut. Sci.* 1 (2) (2010) 47–59.
- [41] M. Rasool, P. Varalakshmi, Protective effect of *Withania somnifera* root powder in relation to lipid peroxidation, antioxidant status, glycoproteins and bone collagen on adjuvant-induced arthritis in rats, *Fundam. Clin. Pharmacol.* 21 (2) (2007) 157–164.
- [42] J.-H. Hwang, S.-W. Lee, S.-J. Rha, H.-S. Yoon, E.-S. Park, K.-H. Han, S.-J. Kim, Dietary green tea extract improves growth performance, body composition, and stress recovery in the juvenile black rockfish, *Sebastes schlegelii*, *J. Aquacult. Inter.* 21 (3) (2013) 525–538.
- [43] N.N. Gabriel, J. Qiang, X.Y. Ma, J. He, P. Xu, K. Liu, Dietary Aloe vera improves plasma lipid profile, antioxidant, and hepatoprotective enzyme activities in GIFT-tilapia (*Oreochromis niloticus*) after *Streptococcus iniae* challenge, *Fish Physiol. Biochem.* 41 (5) (2015) 1321–1332.
- [44] A.D. Talpur, M. Ikhwannuddin, Dietary effects of garlic (*Allium sativum*) on haemato-immunological parameters, survival, growth, and disease resistance against *Vibrio harveyi* infection in Asian sea bass, *Lates calcarifer* (Bloch), *Aquaculture* 364–365 (2012) 6–12.
- [45] S.H. Cho, S.-M. Lee, B.H. Park, S.-C. Ji, J. Lee, J. Bae, S.-Y. Oh, Effect of dietary inclusion of various sources of green tea on growth, body composition and blood chemistry of the juvenile olive flounder, *Paralichthys olivaceus*, *Fish Physiol. Biochem.* 33 (1) (2007) 49–57.
- [46] A. Pérez-Jiménez, H. Peres, V.C. Rubio, A. Oliva-Teles, Effects of diet supplementation with white tea and methionine on lipid metabolism of gilthead sea bream juveniles (*Sparus aurata*), *Fish Physiol. Biochem.* 39 (3) (2013) 661–670.
- [47] B. Cesar, P. Aptekmann, P. Araujo, C. Viagre, C. Maranhao, Orange juice decreases low density in hypercholesterolemic subjects and lipid transfer to high-density lipoprotein in normal and hypercholesterolemic subjects, *Nutr. Res.* 30 (2010) 689–694.
- [48] M. Mackness, C. Abbott, S. Arrol, P. Durrington, The role of high-density lipoprotein and lipid-soluble antioxidant vitamins in inhibiting low-density lipoprotein oxidation, *Biochem. J.* 294 (3) (1993) 829–834.
- [49] S. Hamed, M. Rezaian, T.S. Shomali, Histological changes of small intestinal mucosa of cocks due to sunflower meal single feeding, *J. Am. J. Anim. Vet. Sci.* China 6 (4) (2011) 171–175.
- [50] S. Nordrum, A. Bakke-McKellep, Å. Krogdahl, R. Buddington, Effects of soybean meal and salinity on intestinal transport of nutrients in Atlantic salmon (*Salmo salar* L.) and rainbow trout (*Oncorhynchus mykiss*), *J. Comp. Biochem. Physiol. Part B Biochem. Mol. Biol. Evol.* 125 (3) (2000) 317–335.
- [51] A. Bakke, D. Tashjian, C. Wang, S. Lee, S. Bai, S. Hung, Competition between selenomethionine and methionine absorption in the intestinal tract of green sturgeon (*Acipenser medirostris*), *J. Aquatic Toxicol.* 96 (1) (2010) 62–69.
- [52] L.T. Chwen, H.L. Foo, N.T. Thanh, D. Choe, Growth performance, plasma fatty acids, villous height and crypt depth of preweaning piglets fed with medium chain triacylglycerol, *J. Asian Austr. J. Anim. Sci.* 26 (5) (2013) 700.
- [53] Y.S. Kim, S.B. Ho, Intestinal goblet cells and mucins in health and disease: recent insights and progress, *J. Current Gastroenterol. Rep.* 12 (5) (2010) 319–330.
- [54] M.R.F. Machado, H. de Oliveira Souza, V.L. de Souza, A. de Azevedo, R. Goitein, A. D. Nobre, Morphological and anatomical characterization of the digestive tract of *Centropomus parallelus* and *Centropomus undecimalis*, *J. Acta Scientiarum. Biol. Sci.* 35 (4) (2013) 467–474.
- [55] P. Dharmani, V. Srivastava, V. Kissoon-Singh, K. Chadee, Role of intestinal mucins in innate host defense mechanisms against pathogens, *J. Innate Immun.* 1 (2) (2009) 123–135.
- [56] N.S. Nasruddin, M.N.A. Azmai, A. Ismail, M.Z. Saad, H.M. Daud, S.Z. Zulkifli, Histological features of the gastrointestinal tract of wild Indonesian shortfin Eel, *Anguilla bicolor bicolor* (McClelland, 1844), captured in peninsular Malaysia, *Scientific World J.* 2014 (2014).
- [57] D.A. Vieira-Lopes, N.L. Pinheiro, A. Sales, A. Ventura, F.G. Araújo, I.D. Gomes, A. A. Nascimento, Immunohistochemical study of the digestive tract of *Oligosarcus hepsetus*, *J. World Journal Gastroenterol.* 19 (12) (2013) 1919.
- [58] L. Wilkes, L. Leaver, Effects of different source and level of protein in feed mixtures on liver and intestine histology of the common carp (*Cyprinus carpio*, Linnaeus, 1758), *J. Develop.* 5 (2009) 77.
- [59] V. Poleksić, B. Rašković, Z. Marković, Z. Dulić, M. Stanković, I. Živić, N. Lakić, Effects of different dietary protein sources on intestine and liver morphology of carp yearlings, in: *Proceedings of the 3rd Serbian Congress for Microscopy*, Belgrade, Serbia, Serbian Microscopy Society, 2007, pp. 237–238.
- [60] Z. Zakęs, A. Kowalska, K. Demska-Zakęs, G. Jeney, Z. Jeney, Effect of two medicinal herbs (*Astragalus radix* and *Lonicera japonica*) on the growth performance and body composition of juvenile pikeperch [*Sander lucioperca* (L.)], *J. Aquacult. Res.* 39 (11) (2008) 1149–1160.
- [61] M. Kamali Najafabad, M.R. Imanpoor, V. Taghizadeh, A. Alishahi, Effect of dietary chitosan on growth performance, hematological parameters, intestinal histology and stress resistance of Caspian kutum (*Rutilus frisii kutum* Kamenskii, 1901) fingerlings, *Fish Physiol. Biochem.* 42 (4) (2016) 1063–1071.
- [62] Y. Chen, X. Zhu, Y. Yang, D. Han, J. Jin, S. Xie, Effect of dietary chitosan on growth performance, haematology, immune response, intestine morphology, intestine microbiota and disease resistance in gibel carp (*Carassius auratus gibelio*), *Aquacult. Nutr.* 20 (5) (2014) 532–546.
- [63] S. Torrecillas, A. Makol, M.B. Betancor, D. Montero, M.J. Caballero, J. Sweetman, M. Izquierdo, Enhanced intestinal epithelial barrier health status on European sea bass (*Dicentrarchus labrax*) fed mannan oligosaccharides, *Fish Shellfish Immunol.* 34 (6) (2013) 1485–1495.
- [64] K.A. Denji, M.R. Mansour, R. Akrami, S. Ghobadi, S. Jafarpour, S. Mirbeygi, Effect of dietary prebiotic mannan oligosaccharide (MOS) on growth performance, intestinal microflora, body composition, haematological and blood serum biochemical parameters of rainbow trout (*Oncorhynchus mykiss*) juveniles, *J. Fish. Aquatic Sci.* 10 (4) (2015) 255.
- [65] M. Heidarieh, A.R. Mirvaghefi, A. Sepahi, N. Sheikhzadeh, A.A. Shahbazfar, M. Akbari, Effects of dietary *Aloe vera* on growth performance, skin and gastrointestinal morphology in rainbow trout (*Oncorhynchus mykiss*), *J. Turk. Fish. Aquatic Sci.* 13 (2) (2013) 367–373.
- [66] E.J. Nya, B. Austin, Development of immunity in rainbow trout (*Oncorhynchus mykiss*, Walbaum) to *Aeromonas hydrophila* after the dietary application of garlic, *J. Fish Shellfish Immunol.* 30 (3) (2011) 845–850.

- [67] D.P. Anderson, Immunostimulants, adjuvants, and vaccine carriers in fish: applications to aquaculture, *Annu. Rev. Fish Dis.* 2 (1992) 281–307.
- [68] A. Gupta, S. Singh, Evaluation of anti-inflammatory effect of *Withania somnifera* root on collagen-induced arthritis in rats, *J. Pharm. Biol.* 52 (3) (2014) 308–320.
- [69] A. Grover, A. Shandilya, A. Punetha, V.S. Bisaria, D. Sundar, Inhibition of the NEMO/I κ B association complex formation, a novel mechanism associated with the NF- κ B activation suppression by *Withania somnifera*'s key metabolite withaferin A, *BMC Genom. BioMed. Central* (2010) 1–11.
- [70] E.R. Verrier, C. Langevin, A. Benmansour, P. Boudinot, Early antiviral response and virus-induced genes in fish, *Dev. Comp. Immunol.* 35 (12) (2011) 1204–1214.
- [71] J.P. Mizgerd, M.R. Spieker, C.M. Doerschuk, Early response cytokines and innate immunity: essential roles for TNF receptor 1 and type I IL-1 receptor during *Escherichia coli* pneumonia in mice, *J. Immunol.* 166 (6) (2001) 4042–4048.
- [72] C.-Y. Pan, T.-C. Huang, Y.-D. Wang, Y.-C. Yeh, C.-F. Hui, J.-Y. Chen, Oral administration of recombinant epinecidin-1 protected grouper (*Epinephelus coioides*) and zebrafish (*Danio rerio*) from *Vibrio vulnificus* infection and enhanced immune-related gene expressions, *J. Fish Shellfish Immunol.* 32 (6) (2012) 947–957.
- [73] F. Zanuzzo, E. Urbinati, M. Rise, J. Hall, G. Nash, A. Gamperl, *Aeromonas salmonicida* induced immune gene expression in *Aloe vera* fed steelhead trout, *Oncorhynchus mykiss* (Walbaum), *Aquaculture* 435 (2015) 1–9.
- [74] L. Davis, G. Kuttan, Effect of *Withania somnifera* on cytokine production in nol and cyclophosphamide treated mice, *Immunopharmacol. Immunotoxicol.* 21 (4) (1999) 695–703.
- [75] Z. Li, X. Liu, D. Cao, Y. Liu, X. Wang, J. Liu, Z. Wang, Q. Zhang, H. Yu, Novel functions of interleukin-1 beta in spotted Halibut (*Verasper variegatus*) at its embryonic and metamorphic stages, *J. Ocean Univ. China* 17 (5) (2018) 1197–1208.
- [76] Y. Wang, Q. Wang, P. Baoprasertkul, E. Peatman, Z. Liu, Genomic organization, gene duplication, and expression analysis of interleukin-1 β in channel catfish (*Ictalurus punctatus*), *Mol. Immunol.* 43 (10) (2006) 1653–1664.
- [77] C. Dong, TH17 cells in development: an updated view of their molecular identity and genetic programming, *Nat. Rev. Immunol.* 8 (5) (2008) 337–348.
- [78] C.-N. Zhang, J.-L. Zhang, W.-B. Liu, Q.-J. Wu, X.-C. Gao, H.-T. Ren, Cloning, characterization and mRNA expression of interleukin-6 in blunt snout bream (*Megalobrama amblycephala*), *Fish Shellfish Immunol.* 54 (2016) 639–647.
- [79] N.O. Harun, T. Wang, C.J. Secombes, Gene expression profiling in naïve and vaccinated rainbow trout after *Yersinia ruckeri* infection: insights into the mechanisms of protection seen in vaccinated fish, *Vaccine* 29 (26) (2011) 4388–4399.
- [80] H.-H. Chen, H.-T. Lin, Y.-F. Fong, J.H.-Y. Lin, The bioactivity of teleost IL-6: IL-6 protein in orange-spotted grouper (*Epinephelus coioides*) induces Th2 cell differentiation pathway and antibody production, *Dev. Comp. Immunol.* 38 (2) (2012) 285–294.
- [81] M. Kumari, R.á. Gupta, In vitro antibacterial effect of *Withania somnifera* root extract on *Escherichia coli*, *Vet. World* 8 (1) (2015) 57.
- [82] M.E.-S. El-Boshy, O.M. Abdalla, A. Risha, F. Moustafa, Effect of *Withania somnifera* extracts on some selective biochemical, hematological, and immunological parameters in Guinea pigs experimental infected with *E. coli*, *ISRN Vet. Sci.* 2013 (2013).
- [83] R. Tiwari, S. Chakraborty, M. Saminathan, K. Dhama, S.V. Singh, Ashwagandha (*Withania somnifera*): role in safeguarding health, immunomodulatory effects, combating infections and therapeutic applications: a review, *J. Biol. Sci.* 14 (2) (2014) 77–94.
- [84] M. Serafini, I. Peluso, A. Raguzzini, Flavonoids as anti-inflammatory agents, *Proc. Nutr. Soc.* 69 (3) (2010) 273–278.
- [85] S. Reuter, S.C. Gupta, M.M. Chaturvedi, B.B. Aggarwal, Oxidative stress, inflammation, and cancer: how are they linked? *J. Free Rad. Biol. Med. Glob. Surv.* 49 (11) (2010) 1603–1616.
- [86] E.-J. Park, K. Park, Oxidative stress and pro-inflammatory responses induced by silica nanoparticles in vivo and in vitro, *J. Toxicol. Lett.* 184 (1) (2009) 18–25.
- [87] NRC, Nutrient Requirements of Fish and Shrimp, National Academies Press, 2011.
- [88] E. Zahran, W. Awadin, E. Risha, A.A. Khaled, T. Wang, Dietary supplementation of *Chlorella vulgaris* ameliorates chronic sodium arsenite toxicity in Nile tilapia *Oreochromis niloticus* as revealed by histopathological, biochemical and immune gene expression analysis, *Fish Sci.* 85 (1) (2019) 199–215.
- [89] S. Elbahnaswy, G.E. Elshopakey, Differential gene expression and immune response of Nile tilapia (*Oreochromis niloticus*) challenged intraperitoneally with *Photobacterium damsela* and *Aeromonas hydrophila* demonstrating immunosuppression, *J. Aquacult.* (2020) 735364.